Manduri Kushal Virendra Chowdary

Hyderabad github.com/kushal1989 linkedin.com/in/kushal-manduri

Education

Bachelor of Technology - Artificial Intelligence and Data Science, 2021 – 2025
Vignan Institute of Technology and Science Hyderabad, India

Skills

- Python
- Power BI
- SQL
- Machine Learning
- Exploratory Data Analysis

Project

Swiggy Delivery Time Prediction
- Built a delivery time prediction system using machine learning models on a dataset of 41K+ records and 20+ features, enabling accurate real-world predictions.
- Performed data preprocessing and feature engineering, handling missing values and encoding 9+ categorical features to improve data quality.
- Applied feature selection techniques (Correlation, SelectKBest, Mutual Information) to reduce dimensionality to ~11 key features, enhancing model performance.
- Trained and evaluated models including Random Forest and XGBoost, achieving an R^2 score of ~0.72 on test data while minimizing overfitting.
- Deployed the model using Streamlit, implementing a real-time prediction pipeline with user input handling and feature scaling.

AI-Based Real-Time Face Monitoring System for Online Proctoring
- Developed a real-time face monitoring system using Python, OpenCV, and MediaPipe to analyze live webcam input for online proctoring.
- Implemented face detection and tracking to identify scenarios such as *no face, multiple faces, and normal presence* in real time.
- Designed rule-based alert mechanisms with dynamic on-screen status updates to detect abnormal behavior and enhance exam integrity.
- Optimized system for real-time performance with minimal latency during continuous video processing.

Engineering College Placement Analytics Using Data Driven Insights
- Scraped large-scale real-world data from dynamic web pages using Selenium, extracting thousands of records efficiently.
- Cleaned and preprocessed data using Pandas, handling missing and inconsistent values (~25–30% improvement in data quality).
- Conducted Exploratory Data Analysis (EDA) on datasets with 10+ features to identify patterns, correlations, and outliers.
- Developed 10+ visualizations (histograms, box plots, heatmaps, scatter plots) using Matplotlib and Seaborn to derive actionable insights.
- Designed an end-to-end data pipeline (Scraping → Cleaning → Analysis → Visualization), improving analysis efficiency by ~40%.

Bike Store Sales & Performance Analysis Using SQL
- Designed and analyzed a relational database with multiple interconnected tables (customers, orders, products, stores).
- Performed advanced SQL queries using JOINs, GROUP BY, and HAVING to extract business insights.
- Calculated key metrics such as total revenue, average order value (AOV), and monthly sales trends.
- Identified top customers, best-selling products, and high-performing stores/staff using aggregation queries.
- Conducted customer segmentation and repeat customer analysis to understand purchasing behavior.